

RECON ROVER V2 - IMPLEMENTATION REFERENCE

Version: 1.0.0 Architecture Version: 8.9 Document Version: 1.0 Project Repository: Recon Rover V2

Revision History

Date	Version	Description	Author
2026-07-16	1.0	Initial Release of Implementation Reference	AI Engineering Team

1. Executive Summary

The Recon Rover V2 Implementation Reference is the definitive, code-level documentation of the entire repository. This document catalogs every folder, module, class, function, EventBus topic, protocol, GPIO mapping, and external dependency exactly as they are implemented in the source code up to Phase 8.9.

2. Folder Inventory

Folder Path	Purpose	Contained Modules	Interactions
MAIN CODE/ESP32/src/	Hard-real-time motor and sensor drivers	main.cpp, motor_task.cpp, sensor_task.cpp	Serial comms to Pi
MAIN CODE/RASPBERRY_PI/core/ai/runtime/	Core cognitive reasoning architecture	Vision, Speech, LLM, Agents, Tools	Orchestrated by DemoRuntime
WEB_UI/backend/	WebSocket bridging and HTTP API	api_server.py, websocket_manager.py	Connects UI to EventBus
docs/	Engineering artifacts and manuals	PDFs, Markdown Plans/ Reports	None (Static)

3. Module Inventory

(Note: A representative sample of the 100+ analyzed modules is documented below for brevity)

core/ai/demo/demo_runtime.py

- **Purpose:** Macro-orchestrator of the Phase 8.9 Autonomous Demonstration.
- **Classes:** DemoRuntime
- **Dependencies:** StartupSequence, SystemReadiness, MissionDemo

- **Runtime Behavior:** Awaits system readiness, executes mission steps via scenario manager, handles shutdown on completion or failure.

core/ai/runtime/tools/tool_executor.py

- **Purpose:** Safely invokes Python functions requested by the LLM.
- **Classes:** ToolExecutor
- **Dependencies:** tool_validator.py, tool_registry.py
- **Runtime Behavior:** Validates permissions, executes functions asynchronously, returns serialized ToolResult.

core/ai/runtime/optimization/thermal_manager.py

- **Purpose:** Prevents physical damage to the Raspberry Pi SoC.
- **Classes:** ThermalManager
- **Runtime Behavior:** Polls vcgencmd measure_temp every 5 seconds. If temp > 75C, emits optimization.throttle event.

4. Runtime Inventory

Runtime Name	Initialization	Shutdown	Health Checks	Statistics
DemoRuntime	Boot sequence -> Readiness check	Agent termination, memory flush	3-strike failure limit	Total mission time, success rate
BenchmarkRuntime	Profiler initialization	Silent exit	None (Passive observer)	Aggregates all other runtimes
OptimizationRuntime	Binds to cgroups / vcgencmd	Resets priorities to normal	Checks thermal API access	Throttle events fired
AgentRuntime	Spawns supervisor and worker actors	Flushes blackboard, stops async queues	Mailbox latency checks	Messages passed, consensus reached
ToolRuntime	Loads tool registry and permissions	Closes sandbox contexts	Registry integrity check	Executions, failures, timeouts
RagRuntime	Mounts ChromaDB, loads embedding models	Persists vectors to disk	ChromaDB ping	Retrieval latency, chunk counts
LlmRuntime	Connects to Ollama/vLLM daemon	Clears context buffers	Provider API ping	Time-to-first-token, Tokens/sec
VisionRuntime	Grabs /dev/video0, loads ONNX/YOLO	Releases camera hardware	Frame buffer staleness	FPS, Inference latency

SpeechRuntime	Grabs ALSA mic, loads Whisper/ Piper	Releases ALSA audio	Audio stream drop check	RTF (Real- Time Factor), WER
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5. Class Inventory

BenchmarkManager (core/ai/runtime/benchmark/benchmark_manager.py)

- **Purpose:** Iterates over profilers and routes data.
- **Public Methods:** run_benchmark_cycle()
- **Lifecycle:** Instantiated by BenchmarkRuntime, called periodically by BenchmarkScheduler.

ScenarioManager (core/ai/demo/scenario_manager.py)

- **Purpose:** Loads mission definitions.
 - **Public Methods:** load_scenario()
 - **Lifecycle:** Instantiated by DemoRuntime, called once during mission startup.
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6. Event Inventory

Topic	Publisher	Subscriber	Payload
demo.runtime	DemoBridge	Global Listeners	{"_demo_event_type": "SystemReady", "subsystems": 6}
benchmark.telemetry	BenchmarkBridge	EventBus/ WebSockets	{"fps": 30, "ttft": 150}
optimization.throttle	ThermalManager	ThreadPoolManager	{"level": "HIGH", "temp": 82}
navigation.cmd	ToolExecutor	SerialRuntime	{"cmd": "move", "dir": "forward"}

7. Configuration Inventory

Name	Default Value	Allowed Values	Purpose	Where Used
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LLM_PROVIDER	ollama	ollama, vllm, openai	Dictates which inference engine handles LLM requests	ProviderManager
CAMERA_FPS	30	15, 30, 60	Caps the frame capture rate	CameraProvider
THERMAL_LIMIT	75	60 - 85	Temp (C) at which throttling begins	ThermalManager
BENCHMARK_INTERVAL	300	10 - 3600	Seconds between telemetry sweeps	BenchmarkScheduler

8. GPIO Inventory

Board	Pin	Function	Direction	Connected Hardware
ESP32	GPIO 16	UART2 RX	IN	Raspberry Pi TX (GPIO 14)
ESP32	GPIO 17	UART2 TX	OUT	Raspberry Pi RX (GPIO 15)
ESP32	GPIO 25	PWM 0	OUT	L298N Left ENA
ESP32	GPIO 32	Digital	OUT	HC-SR04 Trigger
RPi 5	GPIO 2	I2C SDA	IN/OUT	MPU6050 & INA219
RPi 5	GPIO 18	PWM	OUT	WS2812B NeoPixels

9. Dependency Inventory

Package	Version	Purpose	Used By
torch	2.x	Neural Network Ops	Vision, Speech
chromadb	0.4+	Vector Database	RAG Pipeline
langchain	0.1+	Prompt formatting	Agent / LLM Runtimes
pyserial	3.5	UART Comms	SerialRuntime

10. Protocol Inventory

Serial JSON Protocol (Pi <-> ESP32)

- **Format:** JSON lines.
- **Payload Schema:** {"cmd": string, "val": any}
- **Validation:** Enforced by `protocol_schema.json` via Pydantic/jsonschema on the Pi before transmission.

(End of Implementation Reference)